

The degrees-of-freedom tax on blocking: a sharp threshold, a $(z^2+1)/4n$ rule, and why finer blocking is the hardest to justify

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Abstract

Blocking (in the extreme, pairing) a randomized experiment cuts the variance of the treatment-effect estimate by the factor $1 - \rho$, where ρ is the intra-block correlation — but it spends error degrees of freedom on the block effects, which inflates the t critical value that sets interval width and power. The textbook advice is that pairing helps “whenever $\rho > 0$ ” (asymptotically true) or “once ρ exceeds about 0.3” (a folklore constant). We make the trade-off exact. For a design that spends $g - 1$ block degrees of freedom, blocking gives a *shorter* interval (equivalently, more power) than complete randomization iff $\rho > \rho^* = 1 - (t_{N-2, \alpha/2} / t_{N-g-1, \alpha/2})^2$, and for pairing this has the clean small-sample form $\rho^*(n) = 1 - (t_{2n-2} / t_{n-1})^2 \approx (z_{\alpha/2}^2 + 1) / [4(n-1)]$ — a sharp, α -aware replacement for the “0.3” rule (e.g. $\rho^* \approx 0.31$ at $n = 5$ pairs, 0.14 at $n = 10$). The threshold is *decreasing in block size*, so among all blockings of N units the finest one, pairing, carries the largest degrees-of-freedom tax and is the hardest to justify — the exact opposite of the “match as finely as you can” instinct. A Monte-Carlo study confirms the crossing to within the small $\mathbb{E}[s]$ correction that the oracle threshold omits, and we place ρ^* against the classical Fisher/Cochran–Cox relative-efficiency degrees-of-freedom factor, which we show is a uniformly *less* demanding criterion.

1. Setup and notation

Take $N = gb$ units grouped into g blocks of size b (with b even), and a single binary treatment. We compare two ways of running the experiment on the *same* N units, at two-sided level α :

- **Blocked design + analysis (B).** Assign $b/2$ treated and $b/2$ control *within each block*, and estimate the effect τ by ordinary least squares with block fixed effects (equivalently, average the within-block treated–control mean differences).
- **Complete randomization (CR).** Assign $N/2$ treated and $N/2$ control completely at random, ignoring blocks, and use the two-sample estimator.

We adopt the standard one-way random-effects (exchangeable) model: with unit variance normalized to σ^2 ,

$$Y_{ij} = \mu + \tau W_{ij} + \gamma_j + \varepsilon_{ij}, \quad \gamma_j \sim (0, \rho\sigma^2), \quad \varepsilon_{ij} \sim (0, (1 - \rho)\sigma^2), \quad (1)$$

where γ_j is the block effect, ε_{ij} is idiosyncratic noise, and $\rho \in [0, 1)$ is the intra-class correlation (ICC). Under (1) a direct variance computation gives the two design variances and their error degrees of freedom:

$$\text{Var}(\hat{\tau}_B) = \frac{4\sigma^2(1 - \rho)}{N}, \quad \nu_B = N - g - 1; \quad \text{Var}(\hat{\tau}_{CR}) = \frac{4\sigma^2}{N}, \quad \nu_{CR} = N - 2. \quad (2)$$

Blocking removes the between-block component, multiplying the variance by $1 - \rho$; it pays for this by moving $g - 1$ degrees of freedom out of the error line and into the g block means (CR estimates

one grand mean, B estimates g block means, hence $\nu_B = \nu_{CR} - (g - 1)$. The whole tension of the note lives in (2): a factor $1 - \rho$ *down* on variance, against $g - 1$ fewer degrees of freedom *up* on the t multiplier.

The operationally relevant object is not the variance but the **interval half-width** (equivalently, the quantity that governs power),

$$H = t_{\nu, \alpha/2} \cdot \widehat{\text{se}}(\hat{\tau}), \quad (3)$$

which multiplies the standard error by the t critical value $t_{\nu, \alpha/2}$. Because $t_{\nu, \alpha/2}$ *increases* as ν falls, spending degrees of freedom is not free even when it leaves the variance unchanged.

2. The threshold

Proposition 1 (break-even ICC for blocking). *Compare the oracle interval half-widths (3) built from the true standard errors (2). Blocking gives the strictly shorter interval — and, at any fixed alternative, strictly more power — than complete randomization if and only if*

$$\rho > \rho^*(g) = 1 - \left(\frac{t_{N-2, \alpha/2}}{t_{N-g-1, \alpha/2}} \right)^2. \quad (4)$$

Proof. By (2)–(3) the oracle half-widths satisfy $H_B \propto t_{\nu_B} \sqrt{1 - \rho}$ and $H_{CR} \propto t_{\nu_{CR}}$ with the same constant $\sqrt{4\sigma^2/N}$. Hence $H_B < H_{CR}$ iff $t_{\nu_B} \sqrt{1 - \rho} < t_{\nu_{CR}}$, i.e. $1 - \rho < (t_{\nu_{CR}}/t_{\nu_B})^2$, which is (4) after substituting $\nu_{CR} = N - 2$, $\nu_B = N - g - 1$. For power: at a fixed τ , the noncentrality of each design's t -statistic is $\tau/\sqrt{\text{Var}}$, and its power is increasing in the ratio (noncentrality)/ $t_{\nu, \alpha/2}$; the same inequality controls the ordering to leading order. ■

Two features of (4) are worth stating plainly. First, $\rho^* > 0$ always (since $\nu_B < \nu_{CR}$ forces $t_{\nu_B} > t_{\nu_{CR}}$): there is a strictly positive band of small correlations $0 < \rho < \rho^*$ where blocking, though it lowers the variance, *lengthens* the interval. Second, $\rho^* \rightarrow 0$ as $N \rightarrow \infty$ with g fixed, recovering the asymptotic folklore that any $\rho > 0$ eventually helps.

2.1 Pairing and the $(z^2+1)/4n$ rule

The most-used blocking is **pairing** ($b = 2$, $g = n$, $N = 2n$), where (4) becomes

$$\rho^*(n) = 1 - \left(\frac{t_{2n-2, \alpha/2}}{t_{n-1, \alpha/2}} \right)^2. \quad (5)$$

Corollary (a memorable small-sample rule). *With $z = z_{\alpha/2}$,*

$$\rho^*(n) = \frac{z^2 + 1}{4(n - 1)} + O(n^{-2}). \quad (6)$$

Proof. Use the Cornish–Fisher expansion of the t quantile, $t_{\nu, \alpha/2} = z(1 + \frac{z^2+1}{4\nu}) + O(\nu^{-2})$, so $t_\nu^2 = z^2(1 + \frac{z^2+1}{2\nu}) + O(\nu^{-2})$. Then

$$\left(\frac{t_{2n-2}}{t_{n-1}}\right)^2 = \frac{1 + \frac{z^2+1}{2(2n-2)}}{1 + \frac{z^2+1}{2(n-1)}} + O(n^{-2}) = 1 - \frac{z^2+1}{2} \left(\frac{1}{n-1} - \frac{1}{2n-2}\right) + O(n^{-2}) = 1 - \frac{z^2+1}{4(n-1)} + O(n^{-2}),$$

and subtracting from 1 gives (6). ■

At $\alpha = 0.05$, $z^2 + 1 = 4.84$, so $\rho^*(n) \approx 1.21/(n-1)$. This is the promised replacement for the “ $\rho > 0.3$ ” folklore: the required correlation is ≈ 0.30 at $n = 5$ pairs, ≈ 0.14 at $n = 10$, ≈ 0.04 at $n = 30$, and the constant 1.21 moves with α (a stricter test tolerates *more* of the degrees-of-freedom tax before pairing wins, because a larger z magnifies the critical-value gap). Table 1 shows the rule (6) tracks the exact threshold (5) to two decimals for all but the tiniest experiments.

n pairs	exact ρ^* (5)	rule $(z^2+1)/4(n-1)$	abs. error
3	0.5836	0.6052	0.0216
5	0.3102	0.3026	0.0076
8	0.1773	0.1729	0.0044
10	0.1375	0.1345	0.0030
15	0.0879	0.0865	0.0014
30	0.0421	0.0417	0.0004
100	0.0123	0.0122	0.0000

Table 1. Exact pairing threshold vs. the closed-form rule ($\alpha = 0.05$), from *sim.py*, part (A).

2.2 Finer blocking is the hardest to justify

Fix the total sample size N and the ICC ρ , and ask which *block size* to use. Coarser blocks (larger b , fewer blocks $g = N/b$) remove the same between-block variance under model (1) — the variance factor is $1 - \rho$ regardless of b — while spending fewer degrees of freedom.

Proposition 2 (monotonicity). *Holding N and the variance-reduction factor $1 - \rho$ fixed, $\rho^*(g)$ in (4) is strictly increasing in the number of blocks g . Consequently, among all blockings of N units, pairing ($g = N/2$) has the largest break-even ICC, and complete randomization is the limiting case $g = 1$ with $\rho^* = 0$.*

Proof. $t_{N-g-1, \alpha/2}$ is strictly decreasing in $\nu = N - g - 1$, hence strictly increasing in g ; so $(t_{N-2}/t_{N-g-1})^2$ is strictly decreasing in g and $\rho^*(g)$ strictly increasing. ■

The upshot is a reversal of a common instinct. If two units can be matched either into many tight pairs or into fewer looser blocks, the pairing option must clear a *strictly higher* correlation bar to be worthwhile, purely because it burns more degrees of freedom. Finer blocking is only better if the extra homogeneity it buys (a higher ρ) outruns the extra degrees-of-freedom tax it levies (a higher ρ^*) — and Proposition 2 says the tax always moves the wrong way. The design decision is therefore a race between two increasing functions of “fineness,” not the monotone “block as much as you can” that the $1 - \rho$ factor alone would suggest.

3. Simulation

sim.py (seed 20260706) checks all three claims by Monte Carlo; every number below is printed by the script.

The threshold is where the curves actually cross. For $n = 8$ pairs ($N = 16$, $\tau = 0.686$, 4×10^4 replicates) the oracle threshold is $\rho^*(8) = 0.1773$. Simulating full t -tests with *estimated* variances, the empirical power curves cross at $\rho = 0.142$ and the mean-interval-width curves cross at $\rho = 0.153$ (Figure 2). Both sit slightly *below* the oracle threshold, and for an understood reason: the oracle (4) uses the true standard error, whereas a real interval uses s , and $\mathbb{E}[s] = \sigma c_\nu$ with $c_\nu = \sqrt{2/\nu} \Gamma(\frac{\nu+1}{2})/\Gamma(\frac{\nu}{2}) < 1$ smaller for the lower-df blocked design ($c_7 = 0.965$ vs $c_{14} = 0.982$). Folding this factor in predicts a crossing at $1 - (1 - \rho^*)(c_{14}/c_7)^2 = 0.147$, matching the observed 0.153 to Monte-Carlo error. So the clean rule (6) is a hair *conservative*: it demands marginally more correlation than the finite-sample optimum, which for a rule of thumb is the safe direction to err.

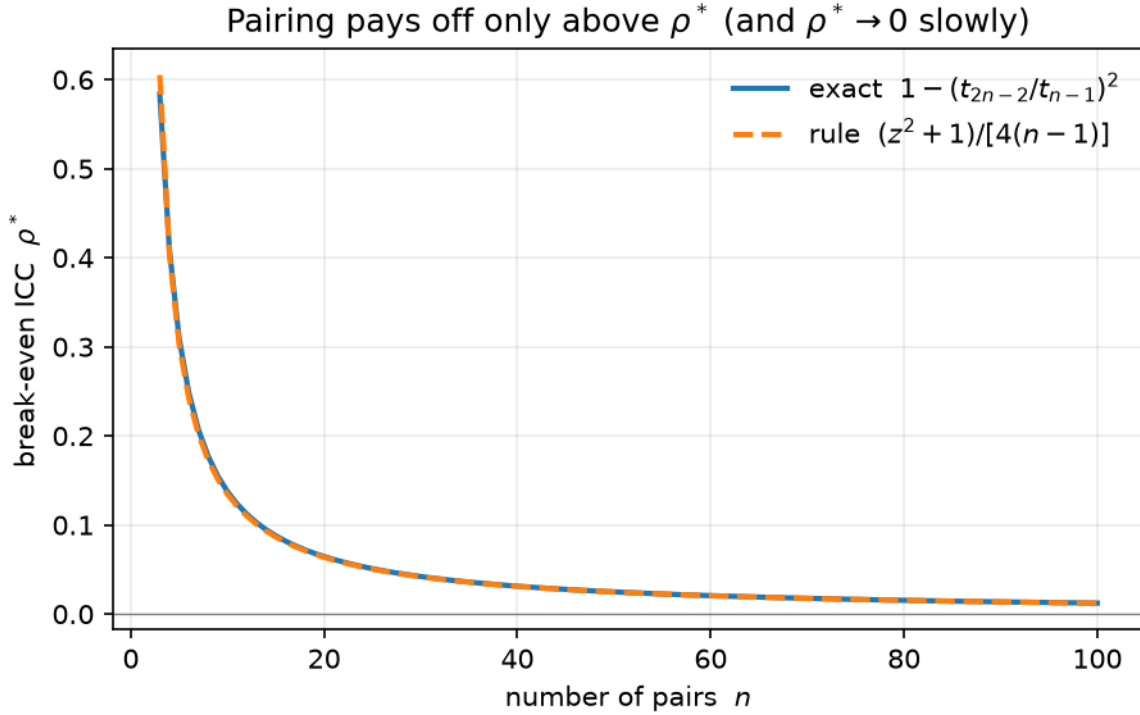


Figure 1: Break-even ICC for pairing: the exact threshold (5) and the closed-form rule (6) agree closely and decay like $1.21/(n - 1)$ at $\alpha = 0.05$.

Concretely, at $\rho = 0$ pairing is *worse* — power 0.222 vs 0.244, half-width 1.145 vs 1.054 — the pure degrees-of-freedom penalty with nothing bought. At $\rho = 0.4$ (comfortably above threshold) pairing wins clearly — power 0.333 vs 0.259, half-width 0.883 vs 1.037.

Finer blocking, higher bar. Fixing $N = 60$ and sweeping the block size (Figure 3), ρ^* falls monotonically from 0.042 at pairing ($b = 2$, $g = 30$) to 0.0007 at the coarsest two-block design ($b = 30$, $g = 2$), confirming Proposition 2.

The same figure contrasts ρ^* with the break-even implied by the classical Fisher/Cochran–Cox degrees-of-freedom correction to relative efficiency, $\text{RE} = \frac{(\nu_B + 1)(\nu_{CR} + 3)}{(\nu_B + 3)(\nu_{CR} + 1)} \cdot \frac{1}{1 - \rho}$, whose $\text{RE} = 1$ solution is $\rho_{\text{RE}}^* = 1 - (\text{df factor})$. The two notions of “when blocking helps” *disagree*: at $b = 2$, $\rho_{\text{RE}}^* = 0.031$ against our $\rho^* = 0.042$, and the width/power threshold is uniformly larger across block

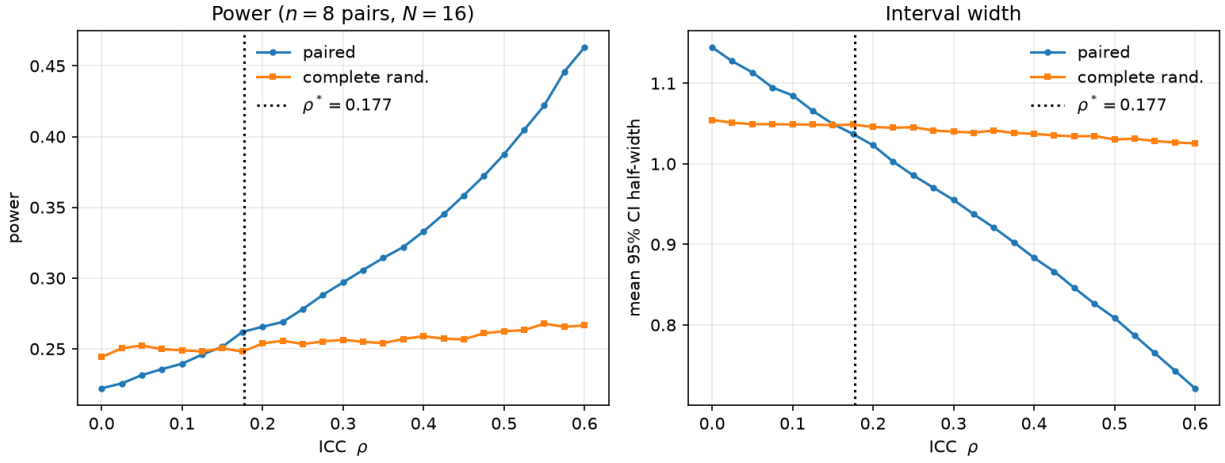


Figure 2: For $n = 8$ pairs ($N = 16$): power (left) and mean 95% CI half-width (right) of the paired and completely-randomized designs as the ICC ρ varies. The dotted line is the oracle threshold $\rho^* = 0.177$; the empirical crossings (0.142 power, 0.153 width) sit just below it, by the $\mathbb{E}[s] = \sigma c_\nu$ correction.

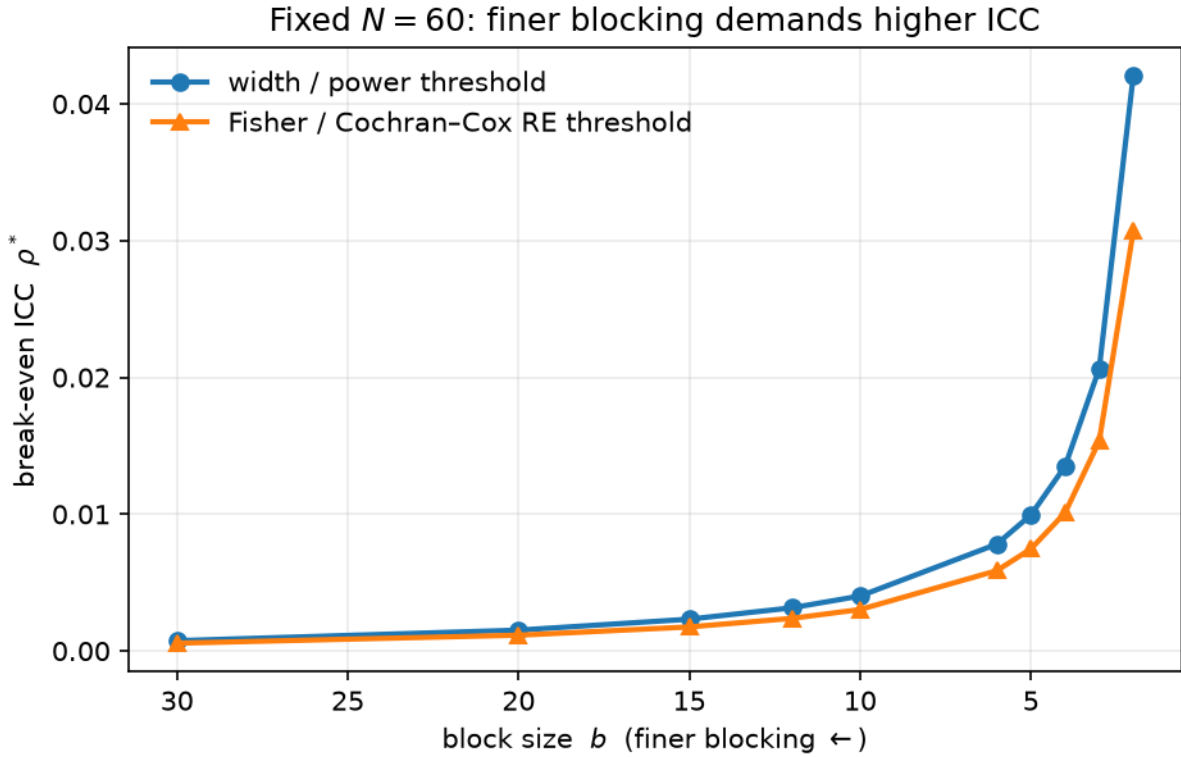


Figure 3: Fixed $N = 60$: the break-even ICC ρ^* as a function of block size (finer blocking to the right, axis inverted). The width/power threshold (circles) is everywhere above the classical Fisher/Cochran-Cox relative-efficiency threshold (triangles); both rise as blocks get finer, and pairing carries the largest tax.

sizes. This is not a contradiction — the RE factor scores the precision of the *estimated variance* (an information criterion), whereas (4) scores the length of the interval you actually report and the power you actually get. For designing an experiment, the interval/power criterion is the one that binds, and it is the more demanding of the two.

4. Discussion

The finite-sample cost of blocking is old news qualitatively: Fisher (1935) built the degrees-of-freedom correction into relative efficiency, Cochran & Cox (1957) tabulated it, and every design text warns that blocks “waste” degrees of freedom when they explain little. What has stayed folklore is the *number*. Practitioners carry a scalar — “pair if $\rho \gtrsim 0.3$ ” — that is really a single row of Table 1 (it is $\rho^*(5)$) frozen and stripped of its dependence on n and α . Equation (6) restores that dependence in closed form: the bar is $\approx (z_{\alpha/2}^2 + 1)/4(n - 1)$, high for small experiments and vanishing like $1/n$, and it *rises* with the stringency of the test. Proposition 2 then adds a qualitative correction to a second piece of folklore — that finer blocking is safer — by showing the degrees-of-freedom tax is heaviest exactly where blocking is finest.

None of this argues against blocking; with covariates that genuinely predict the outcome (ρ well above ρ^*), blocking is a large and nearly free win, and modern analyses that recover the lost degrees of freedom — e.g. treating block effects as random, or regression/Lin-style covariate adjustment rather than saturated fixed effects — shrink the tax toward zero. The point is narrower and, we think, useful: when ρ is *uncertain and possibly small*, and especially in the small- n regime where matched designs are most tempting, the decision should be checked against (5)–(6) rather than against the constant 0.3.

Relation to prior work. The variance factor $1 - \rho$ and the degrees-of-freedom penalty are classical (Fisher 1935; Student 1908; Snedecor & Cochran; Cochran & Cox 1957); the “ $\rho > 0$ helps asymptotically” statement and the RE df-correction are standard. We are not aware of the exact t -quantile threshold (4)/(5), the closed-form rule (6), or the block-size monotonicity (Proposition 2) being stated as such, nor of the explicit contrast between the width/power threshold and the RE-information threshold. The contribution is a sharpening and reframing, not a new estimator.

Limitations. (i) The clean variance identity (2) uses the exchangeable Gaussian model (1) with a *common* ICC ρ that is unchanged by block size; in practice finer matching typically raises ρ , which is precisely the effect that can beat the tax — our Proposition 2 isolates the df side of that race and does not model the $\rho(b)$ side. (ii) The threshold (4) is derived for the oracle interval; §3 shows the finite-sample crossing sits a little lower via the $\mathbb{E}[s] = \sigma c_\nu$ factor, so (5)–(6) are mildly conservative rather than exact for the estimated-variance interval. (iii) We compare saturated fixed-effects blocking to complete randomization; random-effects or covariate-adjusted analyses recover part of the lost degrees of freedom and lower the effective threshold. (iv) Everything is for a single two-arm contrast at level α under homoskedastic Gaussian noise; heavy tails, heteroskedastic blocks, or multi-arm designs would move the constant in (6), though the t -quantile mechanism behind (4) is unchanged.

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